

Kumbara by Sembol

• TESTNET

A self-custodial dollar piggy bank for Türkiye, opened with Face ID.

Live (Stellar TESTNET)

kumbara.sembol.xyz

Repository

github.com/keyboard01/kumbara

Deck date

6 September 2026

Lira savings lose value; the dollar route has a wall in front of it.

Inflation makes lira saving a losing move

Consumer prices in Türkiye rose **31.51% year on year** in August 2026.¹ Dollar-denominated saving is how ordinary households defend themselves.

The crypto route exists but sheds users at every step

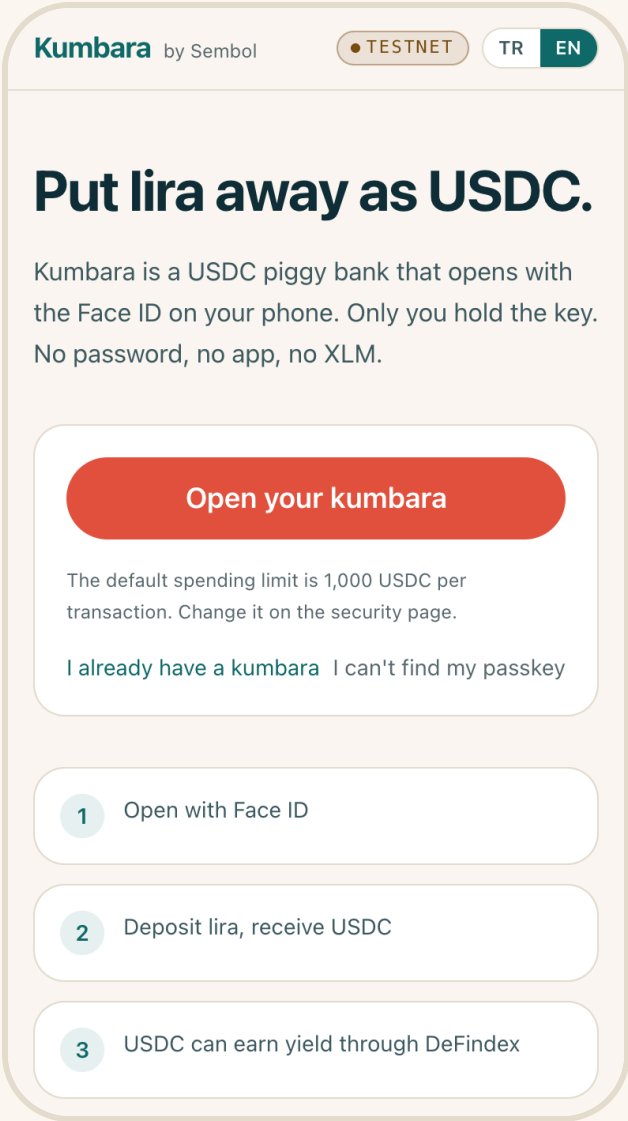
USDC on Stellar settles in seconds for fractions of a cent. Getting there means an extension or app install, twelve words to keep safe, XLM for gas bought somewhere else, and a phone you must never lose.

The people who need hard-currency savings most are served least

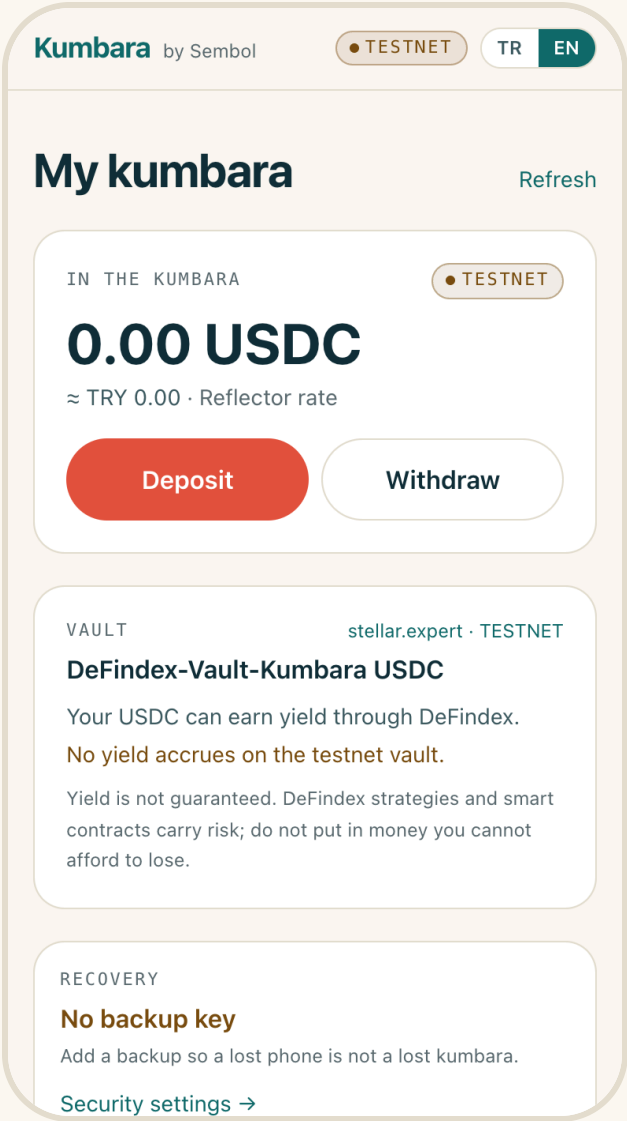
A bank-transfer habit and a phone with Face ID are what people have. That is the whole interface Kumbara asks for: no seed phrase, no XLM, no install.

¹ TÜİK (Turkish Statistical Institute), *Tüketici Fiyat Endeksi, Ağustos 2026*, news bulletin of 3 September 2026: CPI +31.51% against August 2025 (monthly +1.84%). Listed on tuik.gov.tr ("%31,51 (yıl)") and published at veriportali.tuik.gov.tr/tr/press/58290; verified there on 6 September 2026.

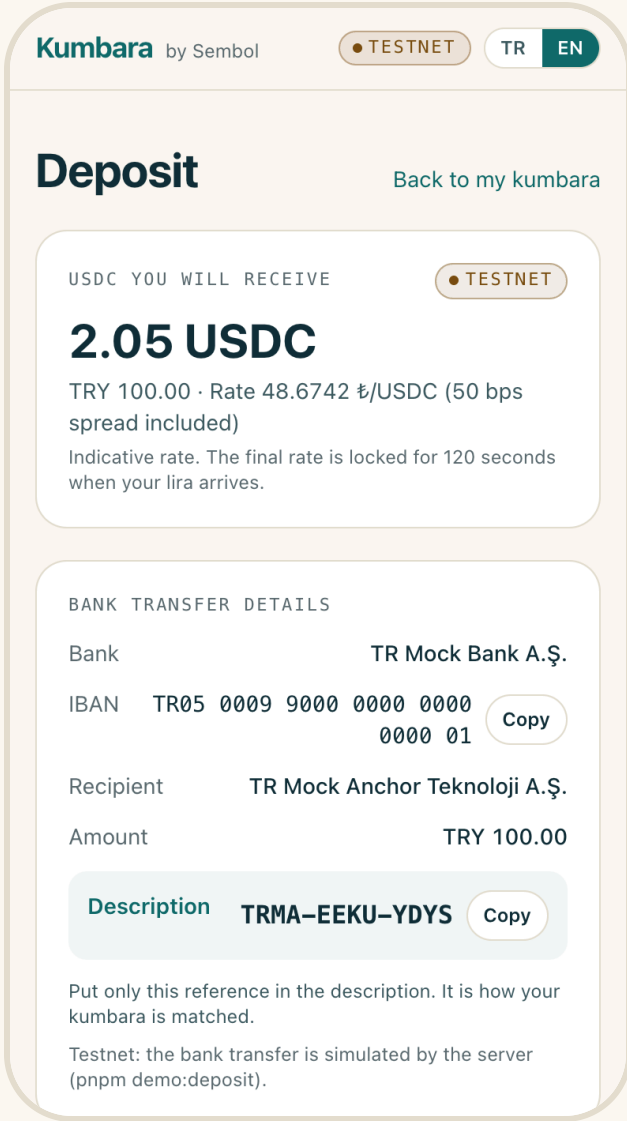
Four screens. Tap to account: ~10 s. No seed phrase, no XLM, no install.



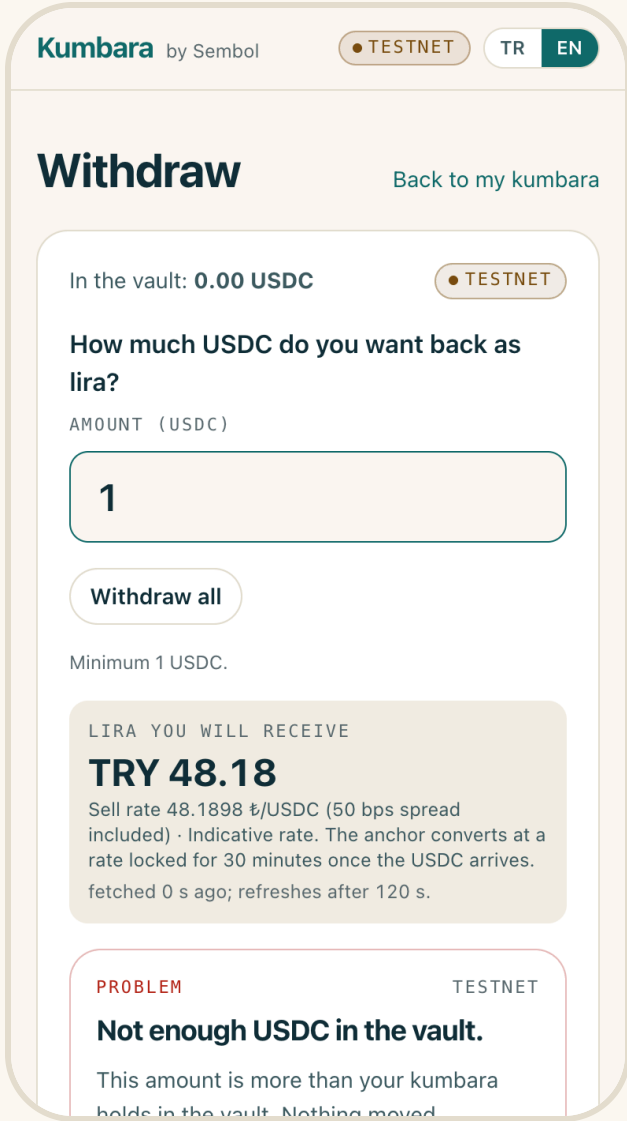
Onboard. One button; Face ID creates the key and the relay deploys an OpenZeppelin smart account. Measured 8–16 s tap to Savings in browser runs.



Savings. USDC in the DeFindex vault with its lira equivalent (Reflector rate), spending limit, recovery. Every figure carries the TESTNET badge.



Deposit. IBAN, recipient and a reference code, exactly like an online bank transfer; the anchor converts lira to USDC.



Withdraw. A sell quote in lira; two Face ID approvals; the anchor pays the visitor's own IBAN.

Scan, tap once, and you have a kumbara before this slide is over.



kumbara.sembol.xyz ● TESTNET

1. Scan the code (it opens the app with this booth's reference).
2. Tap **Kumbaranı aç** and approve with Face ID: an OpenZeppelin smart account is deployed for you; the relay pays the fees.
3. Tap **Yükle**, enter 100, and the presenter plays the bank: about a minute later the USDC is in the DeFindex vault.
4. Tap **Çek** to send it back as lira; the payout reference appears in under a minute.

iPhone: Safari, not private browsing. Android: Chrome with a screen lock. Nothing to install, nothing to write down, no XLM. The counter on the booth screen and kumbara.sembol.xyz/stats move as you go.

HOW IT WORKS

Seven steps, one flow: anchor in front, DeFindex behind, a fee-sponsoring relay underneath.

1

The visitor opens an OpenZeppelin smart account with Face ID; the relay pays the fees.

2

A spending limit (1,000 USDC per transaction) is written on-chain in the background.

3

The anchor issues an IBAN and a reference; the visitor sends lira (simulated by the server on testnet).

4

The anchor converts the lira to USDC and pays an ownerless, pre-authorized bridge account.

5

The bridge account forwards the USDC to the kumbara and closes itself; nobody holds its key.

6

One Face ID puts the USDC into the DeFindex vault (no yield on testnet).

7

A withdrawal runs the path backwards: vault → bridge account → anchor → lira to the IBAN.

The full sequence diagram, every message from the passkey to the payout, is in the appendix (last slide).

Anchors pay classic accounts. Smart accounts are contracts. Ramp payouts could not reach them.

The finding

The anchor's on-ramp rejects a contract (C...) destination and its watcher matches classic payments only: a payout cannot land in a smart account, and a smart account's Soroban transfer is never matched on the way back.

The solution: ownerless, pre-authorized bridge accounts

A throwaway classic account whose only valid transactions are signed before it exists: the **forward** at sequence +1 (USDC to the user's smart account, the exact quoted amount) and the **cleanup** at sequence +2 (trustline dropped, reserves back to the sponsor). The master key is wiped in-process: **no key can sign anything else**. In reverse, the same primitive carries withdrawals to the anchor's treasury with the memo id.

10 / 10

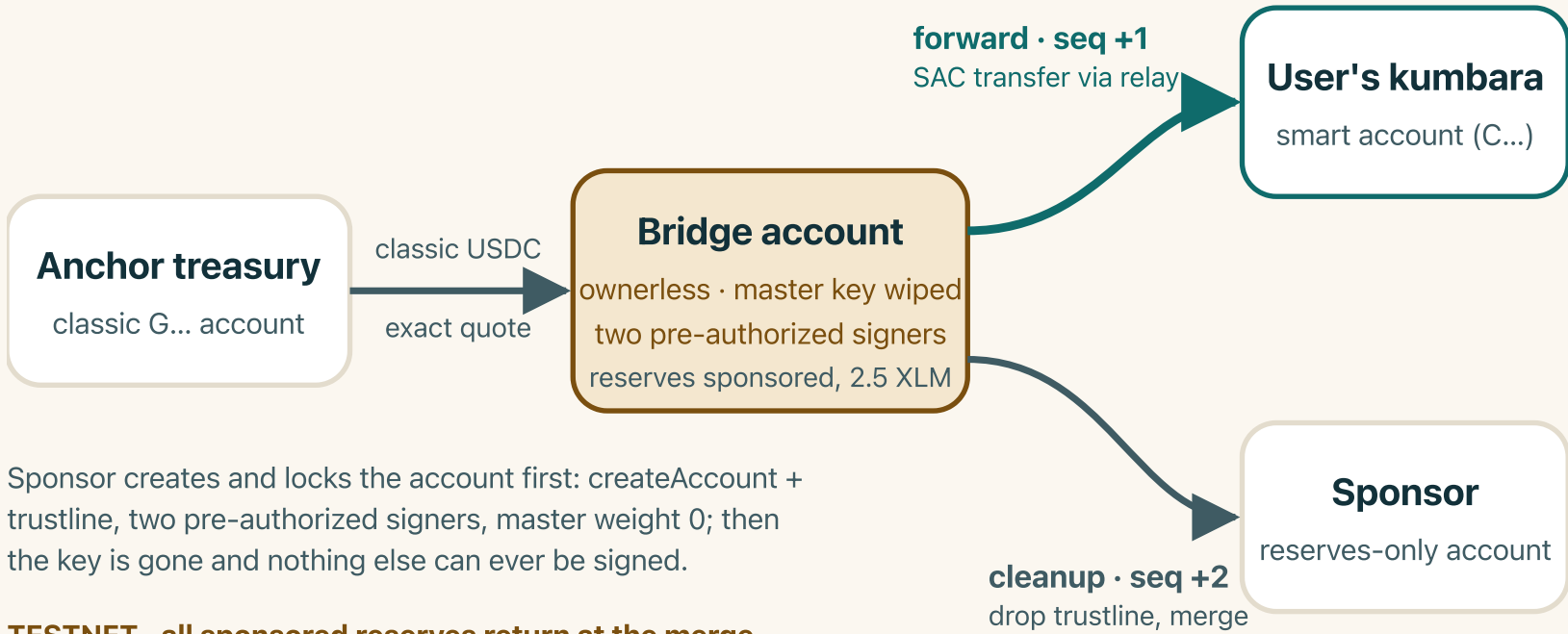
exact amount matches in ten scripted deposits

900

stroops net per deposit for the sponsor; reserves return at cleanup

4

signatures the bridge key ever produces, asserted by a unit test



Sponsor creates and locks the account first: createAccount + trustline, two pre-authorized signers, master weight 0; then the key is gone and nothing else can ever be signed.

TESTNET · all sponsored reserves return at the merge

EVIDENCE (STELLAR TESTNET, STELLAR.EXPERT/EXPLORER/TESTNET/TX/...)	TRANSACTION
Anchor pays the bridge account (classic payment)	778215b3...e3c3a8f1
Pre-authorized forward: USDC lands in the smart account	f580e030...f22c9eab
Pre-authorized cleanup: trustline removed, account merged	6223c61c...7416291d

Reported to the anchor team with the two fixes that would retire the bridge: SAC delivery to contract wallets and Soroban-aware matching on the way back. Full hashes in docs/anchor-notes.md.

Listed building blocks, wired end to end. Kumbara writes no wallet, vault or strategy code.

BUILDING BLOCK	WHAT IT DOES IN THIS APP	ADDRESS (TESTNET, STELLAR.EXPERT)
OpenZeppelin smart accounts + spending-limit policy	Every kumbara is an audited smart account whose only signer is the visitor's passkey; a USDC-scoped rule caps 1,000 USDC per transaction on-chain. Via @sembol/passkey-react and stellar/smart-account-kit.	WASM 1b5f4534...97785a · policy CABXBYJN...2TIP5G
DeFindex vault + hodl strategy	Deposits go through the vault's real invest path into its strategy and withdrawals unwind from it. The strategy is DeFindex's own hodl crate, built unmodified; no yield on testnet, none promised.	vault CAT76PQM...MKSKL · strategy CDFLOE4H...JLDMQN
TR Mock Anchor · SEP-1/6/38 + Partner API	The regulated lira side: IBAN and reference, quotes with a 120 s lock, on-ramps to the bridge account, off-ramps by memo, simulated FAST payouts. Issuer and endpoints read from its stellar.toml.	USDC SAC CBIELTK6...XQDAMA
Fee-sponsoring relay (OpenZeppelin Channels today)	Signed authorization entries go to our server route, which adds the project key and forwards to the relay; the visitor never sees XLM. Per-IP and per-booth caps, sponsor threshold.	channels.openzeppelin.com/testnet
Reflector	The USD/TRY foreign-exchange oracle behind the lira equivalent on the Savings and stats pages (read by simulation, anchor rate as fallback).	CBKGPWGK...4DXMCJZC (mainnet oracle, read only)

Upstream: "Fix wallet creation crash on default Next.js apps" was merged into [stellar/smart-account-kit](#) on 19 August 2026 and shipped in 0.6.2 · [github.com/stellar/smart-account-kit/pull/4](#)

Open-source plan: the repository is MIT-licensed. Kumbara deploys no proprietary contracts: the smart account, WebAuthn verifier and spending-limit policy are OpenZeppelin's audited contracts, the vault and strategy are DeFindex's; the bridge-account logic (`lib/landing`) is MIT in the repo with its tests, and the repository goes public at award time.

Everything on this slide is public, linkable and re-runnable.

Kumbara by Sembol

TESTNET

TR

EN

Kumbara live

live updated 1:17:47 AM

Public, read-only. Testnet: no real lira moves.

Since event start

All time

1

kumbaras opened

TESTNET

1

deposits completed

TESTNET

0

withdrawals completed

TESTNET

TRY 100.00

TRY in

TESTNET

TRY 0.00

TRY out

TESTNET

26.79

USDC

USDC in the vault (live)

TESTNET

kumbara.sembol.xyz/stats: headline numbers with a TESTNET badge, live feed, onboarding curve, timings. The *since event start* / *all time* toggle counts from 19 September 08:00 and shows all-time until then; seed and E2E accounts are excluded by default.

6

real-browser round trips in CI: onboard, deposit, withdraw, booth, stats, failure screens · every push and every 6 h against production

25

failure states, each a TR/EN screen with one action; every flow resumes from stored state

94 · 100 · 100 · 100

Lighthouse mobile on /stats: performance, accessibility, best practices, SEO

E2E · passing · latest production run 6 Sep 2026

GitHub Actions, workflow e2e.yml (repository badge available once the repo is public)

ON-CHAIN EVIDENCE (STELLAR TESTNET)	TRANSACTION
Vault deployed through the DeFindex factory with the hodl strategy attached	dbf0d5b7...6faa8f96
Deposit with invest=true moved USDC into the strategy (invest)	b615e7dc...76b7f981
Withdrawal unwound the shares from the strategy (divest)	466f48d2...07a9f5ca

Recorded fallback demo: docs/demo/round-trip.mp4, the full round trip on kumbara.sembol.xyz, 2:39 with a caption per step.

Non-custodial software in front of a licensed anchor.

- **Kumbara never holds lira, USDC or keys.** Every transfer is authorized by the visitor's own passkey on their own smart account; the bridge accounts have no owner.
- **The licensed anchor is the regulated party:** it holds the fiat, performs KYC and pays out only to the customer's own named IBAN.
- **No payments feature in Türkiye:** no send, no P2P, no merchant flow. Deposit into a vault and withdraw back to your own IBAN, nothing else.
- **Turkish AML rules (MASAK) shape the UX:** named-IBAN in and out, an on-chain per-transaction cap, anchor-side limits and no cash or third-party paths.
- **A Turkish regulatory opinion comes first, self-funded** (legal fees are not an eligible SCF cost); it is obtained during Tranche 1, and the mainnet path stays behind a capped, presenter-only flag until the opinion and the audit are in.

One builder, already shipping in the ecosystem.

Ahmed

Founder · builds and pitches

Instaward grantee (Stellar Türkiye). Author of [@sembol/passkey-react](#), the passkey wallet layer Kumbara runs on, and contributor to [stellar/smart-account-kit](#). Designed and built Kumbara end to end: the app, the bridge accounts, the CI round trips and the evidence in this deck.

Site	ahmed.bet
LinkedIn	linkedin.com/in/ahmed-zm
GitHub	github.com/keyboard01

SHIPPED UPSTREAM

stellar/smart-account-kit #4: "Fix wallet creation crash on default Next.js apps", merged 19 August 2026, in release 0.6.2.

PUBLISHED

@sembol/passkey-react on npm: passkey creation and signing, recovery, spending-limit policy UI. Every screen in this deck runs on it.

FUNDED

Instawards grantee. Kumbara is the consumer proof for that work, built for this submission.

Groundwork, then testnet-complete and secured, then mainnet. Every line has a cost and a check.

DELIVERABLE	HOW REVIEWERS VERIFY	USD
TRANCHE 1 · GROUNDWORK · ≤ 60 DAYS		SUBTOTAL [\$14,000]
Anchor shortlist evaluated against SEP readiness (1, 6, 10, 12, 24, 38); one selected; LOI signed or a documented go/no-go	evaluation doc, LOI or go/no-go memo in the repo	[\$3,000]
Audit Bank application submitted with scope: policy install, bridge accounts, server path	submission confirmation and scope document	[\$1,000]
Relay budgets and allowlists: per-project caps, contract allowlist, refusal alerts	config in repo; a refused call's log and tx	[\$6,000]
Hosted (Vercel + Turso) and self-host (Fly + SQLite) deployment docs with scripted setup	docs URLs; a fresh deploy from the docs	[\$4,000]
TRANCHE 2 · TESTNET-COMPLETE AND SECURED · ≤ 90 DAYS		SUBTOTAL [\$31,000]
Threat model in SDF's STRIDE template format (assets, threat table, mitigations)	docs/security/threat-model.md	[\$5,000]
On-chain monitoring plan in the builders template, implemented for the vault, policy, sponsor and bridge accounts with alert routing	monitoring-plan.md; a live alert on testnet	[\$8,000]
Audit of the policy install, bridge accounts and server path via Audit Bank (our scoping and engineering support only)	audit report link	[\$3,000]
Findings remediated, with regression tests	fix commits, CI green, auditor re-check	[\$8,000]
Mainnet path behind a capped, presenter-only flag: real USDC, enforced budgets	mainnet tx hashes from the capped path	[\$7,000]
TRANCHE 3 · MAINNET LAUNCH AND PUBLIC PILOT · ≤ 90 DAYS		SUBTOTAL [\$32,000]
Mainnet launch: vault, strategy and policy contracts registered; sponsor account and relay budgets live	mainnet contract IDs on stellar.expert	[\$9,000]
Public pilot cohort on the selected anchor's production rails; validation-focused user testing (within the 1/3 exception)	/stats on mainnet, tx hashes, cohort report	[\$12,000]
Metric window: 30-day NAV measurement, public dashboard, footprint registered	NAV dashboard URL; registered IDs	[\$3,000]
Minor: native mobile (Expo) SDK for the passkey layer, published with docs	npm package and docs URL	[\$8,000]
Total ask: the sum of the bracketed lines (Integration Track "Medium" band, \$50k–100k)		[\$77,000]

FINAL-TRANCHE METRIC COMMITMENT

Metric type	DeFindex vault NAV from TRY deposits (USDC in Kumbara's vault position, mainnet)
Threshold	[X] USDC
Measurement window	30 days after the pilot launch, measured over the window, not a snapshot
On-chain footprint, registered at award time	vault contract, strategy contract, spending-limit policy contract, sponsor account (sponsored bridge accounts per CAP-33), app-created smart accounts

The panel ratifies or adjusts the threshold during panel review. Self-generated, wash or gamed activity is excluded; seed and E2E accounts are already tagged out of every number.

SELF-FUNDED · NOT IN THE ASK

Turkish regulatory opinion (legal fees), entity setup (registration costs), the audit fee itself (Audit Bank), and any marketing beyond the validation-focused user testing allowed in tranches 2 and 3. Mainnet waits for the opinion and the audit regardless.

Expansion filter, after the pilot: high inflation plus an existing licensed Stellar anchor. Same app, same bridge, a different anchor in front.

USD cells are bracketed placeholders for Ahmed; subtotals and the total are their sums. Paid in XLM in tranches of 10% on acceptance, 20% at MVP, 30% at testnet, 40% at mainnet: SCF Handbook, Build Award, "Budget & Deliverable Guidelines" → "How Tranche Payments Work" (stellar.gitbook.io/scf-handbook/scf-awards/build-award/budget-and-deliverable-guidelines). Ineligible per the same page: audit costs, marketing beyond the Integration Track exception, legal fees, entity registration.

Every message in the round trip, from the passkey to the payout.

